Multilevel Scattering Center and Deep Feature Fusion Learning Framework for SAR Target Recognition

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***Abstract*— In synthetic aperture radar (SAR) automatic target recognition (ATR), there are mainly two types of methods: the physics-driven model and the data-driven network. The physics-driven model can exploit electromagnetic theory to obtain physical properties, while the data-driven network will extract deep discriminant features of targets. These two types of features represent the target characteristics in the scattering domain and the image domain, respectively. However, the representation discrepancy caused by the different modalities between them hinders the further comprehensive utilization and fusion of both features. In order to take full advantage of physical knowledge and deep discriminant feature for SAR ATR, we propose a new feature fusion learning framework SDF-Net to combine scattering and deep image features. In this work, we treat the attributed scattering centers (ASC) as set-data instead of multiple individual points, which can well mine the topological interaction among scatterers. Then, multiregion multiscale subsets are constructed at both component and target levels. To be specific, the most significant scattering intensity and overall representation in these subsets are exploited successively to learn permutation-invariant scattering features according to a set-oriented deep network. The scattering representations can provide mid-level semantic and structural features that are subsequently fused with the complementary deep image features to yield an end-to-end high-level feature learning framework, which helps enhance the generalization ability of networks especially under complex observation conditions. Extensive experiments on the Moving and Stationary Target Acquisition and Recognition database verify the effectiveness and robustness of the SDF-Net compared against both typical SAR ATR networks and ASC-based models.**

***Index Terms*— Attributed scattering center (ASC), automatic target recognition (ATR), feature fusion learning, limited data, permutation-invariant scattering feature, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is a ground detection system by actively emitting and receiving electromagnetic waves. Due to its high-resolution imaging capability in all-weather day-and-night conditions, SAR automatic target recognition (ATR) plays a significant role in both civil and military reconnaissance fields [1]. However, the unique scattering imaging pattern of SAR makes it much different between SAR data interpretation and visual photograph processing [2], resulting in many challenges of image modalities in practical applications. Therefore, how to make rational use of the scattering characteristics of SAR data in the case of complicated scattering mechanisms and complex image modalities is a crucial problem in SAR ATR, different from infrared and other remote sensing images [3].

In the field of data-driven networks, conventional algorithms are usually carried out in two phases: feature extraction and classification [4]–[6]. The features for ATR include keypoint descriptors, such as scale-invariant feature transform (SIFT) [7] and histogram of oriented gradient (HOG) [8], as well as multidimensional vectors. With the rise of deep learning, superior deep networks [2], [9], [10] have been developed to fit the posterior or joint distribution between data and supervised information in an end-to-end way [11]. This kind of training mode can make more applicable use of extracted features for back-end tasks. Nevertheless, these networks still do not consider the physical signature in SAR data, which leads to exacerbated poor generalization due to the huge discrepancy between diversely imaging modalities, especially under hugely different observation conditions.

In order to cope with the challenges of SAR image modality differences caused by the complex observation conditions, the existing methods focus on designing SAR-data-oriented models or transfer learning. As for the former strategy, there are several methods proposed to alleviate the overfitting problem caused by limited data [2], [12]–[14]. MSAR [13] proposes to solve the sample restriction with metalearning. Label-directed generative adversarial network (LDGAN) [14] solves the shortage in training samples by generating labeled samples, while RotANet [11] aims to solve the rotation angle sensitivity problem by characterizing the rotational pattern of target poses. On the other hand, data augmentation and transfer learning have been widely used to solve other image modalities problems by increasing the diversity of data representations [15]–[17]. The knowledge expanded by augmentation or learned from other domains can assist the network with its current task. However, fine-tuning based on optical remote sensing or even SAR images has only little effect on SAR ATR because of the huge discrepancy between target and source domain [18]. Therefore, relying solely on the image domain cannot extract adequate discriminative features of different image modalities for SAR ATR.

Manuscript received January 25, 2022; revised April 19, 2022; accepted May 7, 2022. Date of publication May 12, 2022; date of current version May 26, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant U20B2067, Grant 62126213, Grant 61806165, and Grant 61790552; and in part by the Natural Science Basic Research Plan in Shaanxi Province of China under Grant 2020JQ-196. *(Corresponding author: Zaidao Wen.)*

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Digital Object Identifier 10.1109/TGRS.2022.3174703

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In view of the unique scattering mechanism in SAR imaging, the specificity attributes in scattering domain should be involved to increase crucial information for SAR ATR with huge image modality discrepancies. According to the coherent property of electromagnetic waves, there are mainly three kinds of physical model to describe the intrinsic attribute of SAR data: the statistical model [19], [20], the polarization model [21]–[23], and the scattering model [3], [24]–[27], respectively. The statistical models describe the statistical property of SAR data as bright and dark spots, and homogeneous and heterogeneous patterns, while the polarization models refer to focus on the distinctions between polarization modes. Different from the above two models, the scattering model presents the imaging process, and electromagnetic echoes of SAR in detail, thus, are able to describe the physical semantic information and represent target microstructures in the frequency domain. This provides a new path to further extract additional scattering features.

In order to utilize the scattering information in SAR data, several models utilizing attributed scattering centers (ASC) have been proposed. Based on multiple previous theories and methods [28]–[30], the parametric ASC model describing scattering coefficients in the imaging process is presented in [31]. This ASC model constructs a function of frequency and aspect angle to describe SAR echoes, which is now extensively used in SAR scattering studies. Jianxiong *et al.* [3] constructed a feature template based on ASCs, and then, different levels of regional features are matched with the template to make a recognition decision, while Ding *et al.* [32] combined the ASC and random projection feature, which is the first one to perform feature-level fusion based on ASC. On the basis of feature fusion, Zhang *et al.* [33] groundbreakingly fused both deep feature and ASC feature for SAR ATR. By transforming ASCs into feature vectors through the bag of visual words (BOVW), FEC concatenates ASC feature and pretrained VGG [34] deep feature. Though the fusion strategy deserves affirmed, the BOVW feature can only present statistical characteristics of ASC, and the two-phase approach hinders the network from further learning scattering features. The multiscale convolutional neural network based on component analysis (CA-MCNN) [35] uses the reconstructed ASC images to emphasize key regions in multiscale convolutional neural network (CNN) features. Though the different types of components are taken into consideration, the CA-MCNN still does not use ASCs directly and ignores multiple types of ASCs that may exist in one single region.

All in all, the huge presentation discrepancy caused by the different modalities between scattering feature and deep feature makes it much more difficult to directly fuse ASC and SAR data at the feature level. The existing ASC-based methods lack the expression of ASC sets at the component level. The closed-loop end-to-end training approach has not been implemented, which results in poor performance in the further extraction of high-level scattering features. On the other hand, though the deep network can learn powerful discriminant features, the demand for both a large amount of supervision information [36] and homogeneous imaging modalities prevents its application in the case of complex observation conditions. How to make comprehensive and appropriate use of scattering features and deep feature remains to be resolved.

In this article, a novel scattering and deep feature fusion network (SDF-Net) is proposed to fuse multilevel scattering center and deep feature effectively. By directly utilizing ASCs to extract learnable scattering features and deep features in an end-to-end manner, SDF-Net performs a fusion strategy at the feature level to further explore the intrinsic properties of targets. The additional semantic and discriminate information is then mined to enhance the learning capability and generalization performance of networks. The main contributions are summarized as follows.

1) A novel feature fusion learning framework based on scattering feature and deep feature is proposed in this article. To this end, we reconsider the ASCs as set-type data rather than individual points, which provides a new path to extract high-level scattering features by capturing the spatial topological structures of the target. The twice scattering feature extraction strategy contributes to exploring the permutation-invariant representations among ASCs on both local component and global target scales. It is worth noting that the scattering feature can be further learned and optimized in the subsequent networks. Due to the semantic representations of microstructures in ASCs, the multilevel scattering feature can improve the interpretability of the framework.
2) Multiregion multiscale electromagnetic scattering feature and deep feature are fused complementary by constructing an end-to-end pseudo-Siamese network. The scattering feature contains physical information of target multilevel structures, while the deep feature contains discriminant representations. Therefore, the Siamese training mode promotes the complementary fusion and comprehensive utilization between the paired features.
3) The proposed framework can characterize the implicit scattering property and essential attributes both in the scattering domain and the image domain, and makes full use of available supervision knowledge to significantly enhance the generalization capability of models with huge image modality discrepancies. In ten types of SAR target recognition tasks, our method achieves 7.98% more accurately than other advanced algorithms with only 5% training samples, which demonstrates the superiority of the proposed SDF-Net.

The remainder of this article is organized as follows. The geometrical theory of diffraction (GTD)-based ASC model and the paradigm of the deep learning model are briefly introduced in Section II. Section III presents the framework of the proposed method in detail. Then, extensive experiments and analysis are discussed in Section IV. Section V summaries the whole article in general.

II. Preliminary Knowledge

In this section, we first briefly introduce the ASC model based on GTD to describe SAR data in the frequency domain. Then, the ordinary paradigm of the deep learning algorithm will be illuminated.

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A. ASC Model

As an extension of physical optics (PO), the GTD [37] indicates that the electromagnetic fields reflected from targets are approximately equivalent to a set of individual scattering centers in high-frequency conditions. According to this theory, the ASC model is constructed in [31]. The ASC model builds a function of frequency and aspect angle to describe the scattering filed as

$$E(f,\phi;\Theta)=\sum_{i=1}^{n}E_i(f,\phi;\Theta_i) \tag{1}$$

where $E(f,\phi;\Theta)$ represents the total scattering filed of n ASC, and f and ϕ are the operating frequency and aspect angle of radar, respectively. Θ_i and E_i are the physical parameter set and the scattering filed of the ith scattering center. For the ith scattering center, its scattering field formula is given as follows:

$$\begin{aligned}E_i(f,\phi;\Theta_i)=\;&A_i\cdot\left(j\frac{f}{f_c}\right)^{\alpha_i}\\&\cdot\exp\left(\frac{-j4\pi f}{c}(x_i\cos\phi+y_i\sin\phi)\right)\\&\cdot\mathrm{sinc}\left(\frac{2\pi \mathrm{f}}{\mathrm{c}}\mathrm{L_i}\sin(\phi-\bar{\phi}_\mathrm{i})\right)\\&\cdot\exp(-2\pi f\gamma_i\sin\phi)\end{aligned} \tag{2}$$

where c is the propagation speed of light and f_c represents the central frequency of radar. $[A_i, x_i, y_i, \alpha_i, L_i, \bar{\phi}_i, \gamma_i]$ together constitute the parameter set Θ_i mentioned above, where A_i, a scalar, is the amplitude of ith scattering field, α_i reflects its frequency dependence, $[x_i, y_i]$ are its geometrical coordinates, L_i and $\bar{\phi}_i$ represent the length and the orientation angle of the ith distributed scattering center, respectively, and γ_i describes its aspect dependence.

Though the ASC model represents the inherent and intrinsic properties of SAR data, it is still difficult to directly use these discrete scatterers for feature learning. The existing applications for ASC models are still at the stage of statistical feature and reconstructed SAR images. In this article, we will explore the appropriate and effective ASC application method to extract important scattering characteristics of SAR targets.

B. Paradigm of Deep Learning Model

In ordinary deep learning-based models, the networks mainly rely on learning the mapping function $\boldsymbol{g}$: $X \rightarrow Y$ from image domain X to label domain Y to achieve accurate target recognition. In order to fit the joint or posterior distribution between samples and supervision information, the networks generally optimize trainable parameters by minimizing the objective function

$$\min_{g\in\mathbf{G}}\sum_{i=1}^{N}\ell(g(x_i),y_i) \tag{3}$$

where x_i and y_i are the ith sample and its corresponding label, N is the total number of targets, and ℓ is the object loss function of networks.

Considering the additional prior knowledge Z provided by physical model $\boldsymbol{f}$: $X \rightarrow Z$ that maps the samples X into physical representations Z, the deep networks containing Z will learn to fit the joint distribution $g(X, Z)$ of image and physical prior. Then, the minimized objective function will be as the form of

$$\min_{g\in\mathbf{G}}\sum_{i=1}^{N}\ell(g(x_i,\,f(x_i)),\,y_i) \tag{4}$$

where $z_i = f(x_i)$ is the prior knowledge of the ith sample x_i.

In order to precisely learn the discriminant feature for SAR ATR, the networks generally require samples of the same modalities or similar distributions for training. However, the generalization degradation problem occurs when the observation condition changes, while the powerful intrinsic prior of the target will be beneficial to the ATR with image modality discrepancies. The proper fusion method between deep networks and physical models will be explored in this article.

III. Multiregion Multiscale Scattering Feature and Deep Feature Fusion Learning Framework

In this section, the detailed structures of the proposed multiregion multiscale scattering feature and deep feature fusion learning framework will be presented. First of all, we improve the ASC extraction algorithm to obtain ASC sets more precisely and effectively. Then, multiregion and multiscale permutation-invariant scattering features are extracted to represent the microstructures and inherent properties of targets. Subsequently, the deep features are extracted through an all-convolutional network. In order to learn complementary supervision information better, we fuse these two kinds of features in a weighted end-to-end way. The fused features, finally, represent the holistic target in both the scattering domain and the image domain for ATR. The overall framework of the proposed SDF-Net is shown in Fig. 1.

A. Multiregion and Multiscale Scattering Feature Extraction Module

Joint optimization of scattering centers is a nonlinear nonconvex parameter estimation problem [38]. The existing ASC extraction methods are mainly conducted in the image domain [39]–[41] or the frequency domain [42], [43]. To address the problem of oversegmentation and memory-consuming, Jing and Zhang [38] proposed an ASC extraction method (AFEM-GA) based on a genetic algorithm and improved relaxation method. AFEM-GA extracts ASCs with float-coded variables [44] in the frequency-aspect domain to improve the accuracy of estimation. Duan *et al.* [45] proposed to represent the $[x_i, y_i]$ of the ith scattering center with coordinates of remaining strongest pixel in the image domain. To improve the efficiency and precision of parameter estimation, we combine the estimation approaches of locations in [45] and that of other parameters in [38], respectively. The ASC extraction method used in this article is summarized in Algorithm 1.

Calculated by the ASC extraction algorithm introduced above, the comparison between T72 in the moving and stationary target acquisition and recognition (MSTAR) database and the corresponding ASC reconstructed image is shown in

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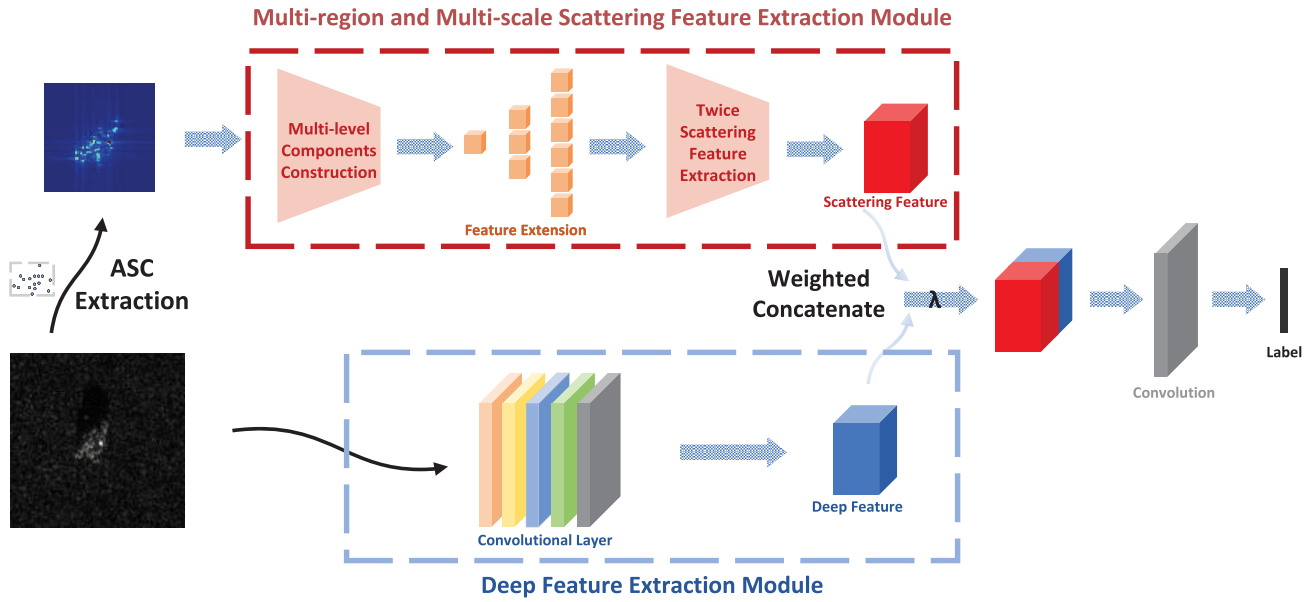


Fig. 1. Overall framework of multiregion multiscale scattering feature and deep feature fusion learning framework.

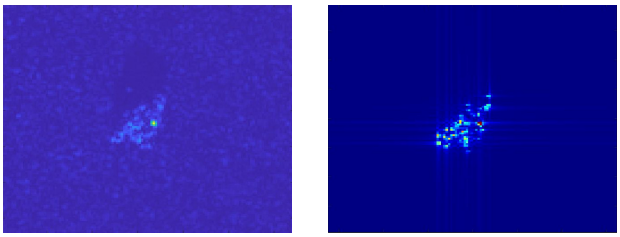
Algorithm 1 Multiple ASCs' Extraction Algorithm

Input: SAR data I(x, y), maximum iteration iterMax, residual criterion resMin.
Output: parameter set Θ.
Initialization: parameter set $\Theta = \text{Zeros}[n, 7]$, SAR scattering field $S_{total} = FFT_{2d}(I(x, y))$ estimated scattering filed $S_{scatter} = \mathbf{0}$, iter $= 1$,i $= 1$,$S_{residual} = S_{total}$
For $(i = 1, i < n, i++)$
While {iter $<$ iterMax & $S_{redisual} >$ resMin}
1. Calculate the residual image $I_{residual}$ $I_{residual} = FFT_{2D}{}^{-1}(S_{residual})$
2. Choose the coordinates of maximum in residual image as $[x_i, y_i] = \text{Coordinate}[\text{Max}(I_{residual})]$, estimate the other parameters$[A_i, \alpha_i, L_i, \bar{\phi}_i, \gamma_i]$ with AFEM_GA. Update the parameter set Θ
3. Utilize the improved relaxation algorithm proposed in AFEM_GA to alleviate the accumulation error
4. Calculate the estimated ASC field and residual scattering filed: $S_{scatter} = E(f, \phi; \Theta)$, $S_{residual} = S_{total} - S_{scatter}$
5. Update the number of iterations iter $=$ iter $+ 1$

End While
End For

Fig. 2. It is obvious that the reconstructed image accurately reproduces the structure of T72, and these ASCs are perfectly located at their correspondent pixel points.

The composition of extracted ASC contains space locations and other physical parameters of target components; such data structures remind us of point cloud data, which represents points in the 3-D space. There are usually 3-D coordinates, colors, and material reflection intensity in typical point cloud data,



(a) (b)

Fig. 2. Comparison between T72 in the MSTAR database and its reconstructed image based on ASCs. (a) T72. (b) Reconstructed T72.

which is pretty similar to ASCs of SAR data. As a collection of 2-D points, ASCs are ought to be treated as a set of items that interact with each other instead of individual points during the feature learning period. However, the existing methods still use ASCs as multiple individual points, which is not conducive to the utilization and feature learning of scattering centers. Inspired by the idea of Deep sets [46], we think that the permutation-invariant property is of great importance to ASCs since ASC sets can represent component-level structures of targets.

We design a multiregion multiscale scattering feature extraction module based on the idea of the set, as shown in Fig. 3. This module mainly consists of three parts: feature extension, multilevel components construction, and the twice permutation-invariant scattering feature extraction module. The detailed structures will be introduced as follows:

First, the multiregion multiscale components will be constructed. Assuming that the number of ASCs extracted from SAR data is N, we use the farthest point sampling (FPS) [47] algorithm to randomly sample N_c ASCs as central points. Considering the huge differences between target structures, such as top-hat, dihedron, and cylinder, as shown in Fig. 3, FPS will sample the central ASCs so that these central points are

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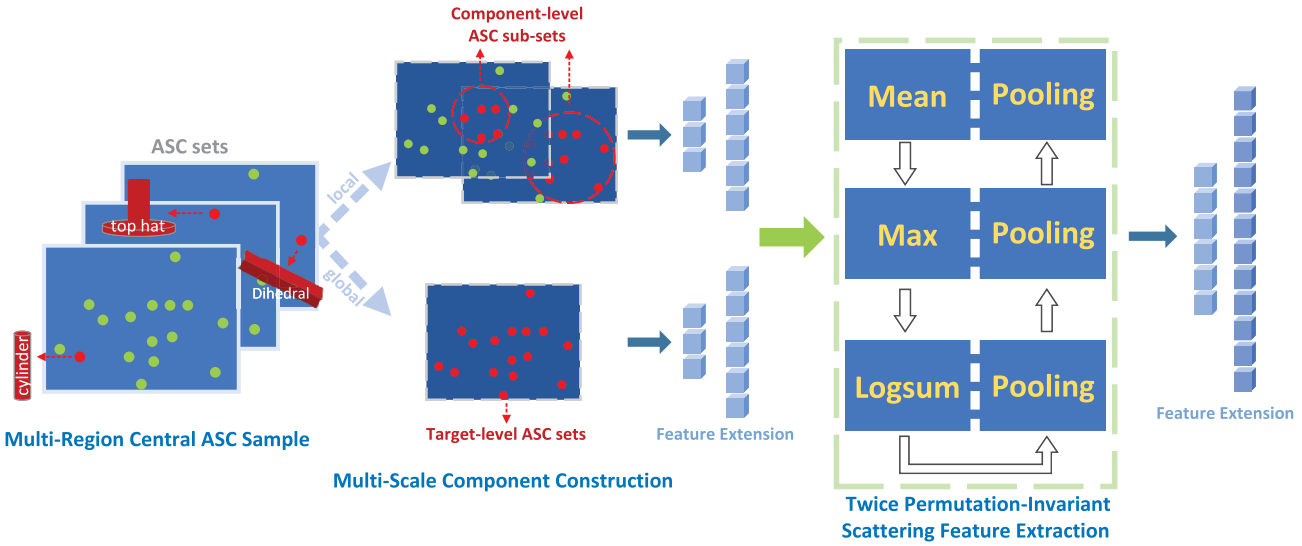
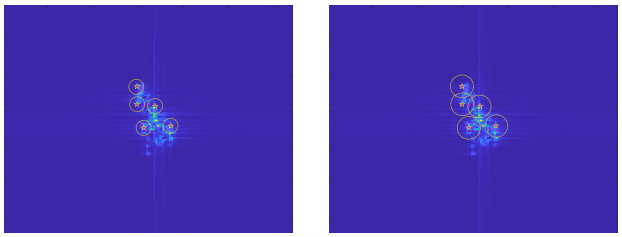


Fig. 3. Multiregion and multiscale scattering feature extraction module.



(a) (b)

Fig. 4. Constructed ASC subsets with N_c and R_i, N_i. (a) $N_c = 5$, $R_i = 1.5$, $N_i = 3$. (b) $N_c = 5$, $R_i = 2$, $N_i = 5$.

as ubiquitous as possible in all spatial locations of the target. Then, the nearest N_i ASCs within radius R_i of each central point are taken as the neighborhood points of it; the sampled ASCs will be resampled to ensure their spatial relations if the number of ASCs within R_i is less than N_i. These central points along with their neighborhood points constitute the multiregion components that describe the structures in different parts of targets, such as bodywork and barrel. Besides, in order to investigate the ASCs in different scales, we further take ASCs within multiple radii R_i around each central point as its multiscale neighborhood points, where $R_i \in [R_1, R_2, \ldots, R_k]$. The number of their neighborhood points is denoted by N_i corresponding to the radius R_i, where $i \in [1, 2, \ldots, k]$. To explore multiscale scattering features in different localities, the numbers N_i of the first $k-1$ neighborhood points of each central ASC and their corresponding radius R_i are set to less than N_c and the target size, respectively. In addition, N_k and R_k are set to be 1 and ∞ to represent all the ASCs of the target to extract global scattering features. Therefore, the local and global ASCs can be combined to together describe the multiregion and multiscale scattering features. The constructed ASC subsets with different scales are shown in Fig. 4.

In order to extend the dimension of ASCs so that the network can learn the scattering feature more comprehensively, the feature extension will transform the ASC sets into higher dimension vectors without destroying their spatial relations. The 1×1 convolution only extends the physical feature dimension; the relative spatial locations of extracted ASCs subsets remain constant. Therefore, the subsequent feature extraction will learn the representations with components' structures.

Finally, the twice permutation-invariant feature extraction module in Fig. 3 is designed to extract scattering features both at the component level and the target level. The twice extraction refers to extracting scattering features of single ASC subsets and all the ASC sets successively. The extracted permutation-invariant scattering features are more suitable for representing the discrete ASC points than traditional deep features. We use global pooling [48] across both local component-level ASC subsets and global target-level ASC sets. In order to obtain the most significant scattering intensity and overall representations of target ASCs, we select three corresponding pooling functions: max pooling, mean pooling, and log-sum pooling, respectively. To be specific, the permutation-invariant feature extraction module is carried out in two steps. First, the pooling functions will conduct in the neighborhoods of each central ASC to obtain three pooling outputs with a size of $[N_c \times 1 \times N_d]$, where N_d is the extended feature dimension. Then, we concatenate these three outputs into local scattering feature maps with a size of $[N_c \times 3 \times N_d]$. Similarly, the global scattering feature maps will be extracted by conducting pooling functions in all the central ASCs. The size of the final local and global scattering feature map is $[3 \times 3 \times N_d]$.

Then, the concatenated feature maps will be fed into another stacked 1×1 convolutional layer to further extract intrinsic and semantic scattering features inside the ASCs. It is worth noting that the scattering features extracted in this module can be further mined and learned in the following feature fusion learning stage.

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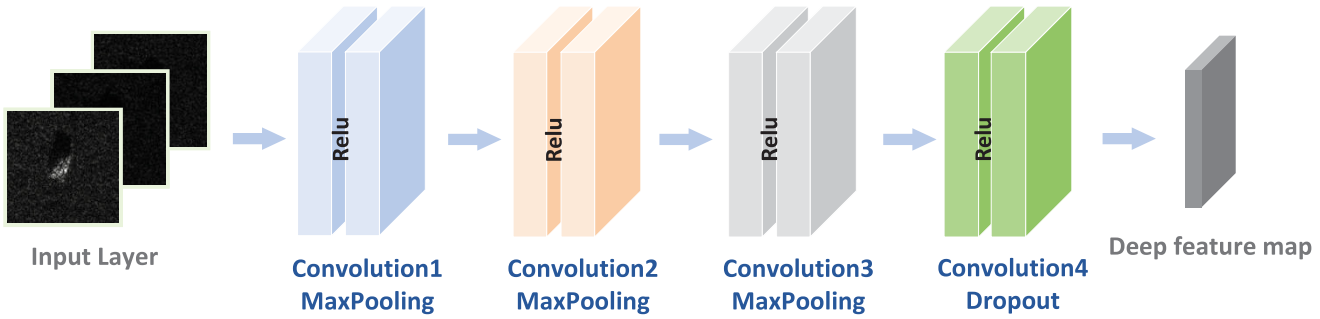


Fig. 5. Deep feature extraction module.

B. Deep Feature Extraction Module

Although the scattering features introduced above are of physical significance, we still need the powerful discriminative capability of deep features to jointly describe the target in SAR ATR. As deep learning develops, plenty of deep networks have been designed for various types of data. However, most of them are constructed on the basis of optical images and require numerous labeled samples for training to well fit the distribution between data and supervision information inside. When it comes to realistic application scenarios with SAR images, too many trainable parameters will lead to overfitting in ordinary optical-image-oriented deep convolutional networks. In view of too many trainable parameters in fully connected (FN) layers, Chen *et al.* [2] proposed an all-convolution network named A-ConvNet to reduce the volume of the network by discarding the FN layers for SAR ATR. Inspired by them, we stack convolutional layers to construct the deep feature extraction module to learn the discriminant deep features of targets. The detailed structure of the network for deep feature learning in this article is shown in Fig. 5.

In the deep feature extraction module, the input SAR data will pass through a stacked network consisting of four convolutional layers to obtain the deep feature maps of the SAR target. It is worth noting that the size of this feature map, $[3 \times 3]$, is exactly the same as the one generated in the scattering feature extraction module, so we can fusion these two kinds of features in a concatenated way efficiently. The deep features extracted in this module represent the contour or key-point features of the target in the image domain, which can be used in the following feature fusion module.

C. Scattering Feature and Deep Feature Fusion

Through the multiregion and multiscale scattering feature extraction module and the deep feature extraction module, SAR data and their corresponding ASC will be both transformed into the form of feature maps. These two kinds of feature maps represent the component-level semantic physical features in the scattering frequency domain and discriminant features in the image domain, respectively.

In order to demonstrate the two kinds of features more precisely, we visualize the outputs of the mentioned two modules in Fig. 6. Each row in the 20-row feature maps

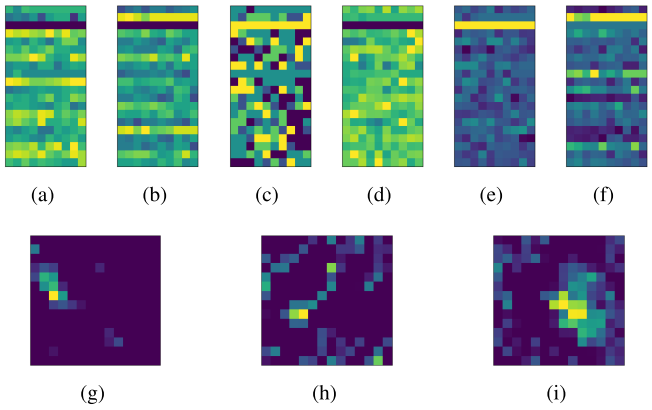


Fig. 6. Feature maps obtained from the scattering feature extraction module and deep feature extraction module. Feature map of (a) x, (b) y, (c) A, (d)–(f) 1×1 convolution module, and (g)–(i) deep feature extraction module.

in Fig. 6(a)–(f) represents a region containing ten individual ASCs, which means that the numbers of central and neighborhood ASCs are 20 and 10, respectively. Fig. 6(a)–(c) shows the x, y, and A physical feature dimensions in ASCs' sets obtained from the output of the sampling and grouping step, and the feature maps in Fig. 6(d)–(f) show the expanded feature dimension in ASC features obtained from the output of the first layer in 1×1 convolution module. The pixels in the third row in Fig. 6(a) are all in dark, which means that the x values of ASCs in this region collectively represented by these pixels are very small. Similarly, the meaning of each pixel in Fig. 6(b)–(f) can be deduced by analogy. For example, it can be seen that the second row in Fig. 6(f) is mostly in yellow, indicating that these ASCs in this feature dimension are of great significance, and this group of ASCs represented by this row will be paid more attention in the following max-pooling section. Both the constructed ASC sets and scattering features can provide more intuitive physical information of component locations, scattering intensity, and aspect angles compared to the deep features. It can be concluded that the multilevel scattering features can improve the interpretability of the framework.

On the other hand, (g)–(i) in Fig. 6 are taken from the third layer of the deep feature extraction module; these feature maps obviously represent the contour and key-point features

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of the SAR target. Therefore, these deep features are of great discriminant, which can be utilized in target recognition.

The scattering features contain structural information and physical properties of targets, while the deep features provide key points and contours of targets. Thus, the scattering features and deep features are complementary and mutually reinforcing for SAR ATR. Therefore, we design a weighted fusion method to fuse these two features efficiently. To be more specific, a weight coefficient λ is designed to concatenate them. For instance, assuming that the scattering and deep features are denoted as f_s and f_d, the sizes of scattering features and deep features obtained in experiments are $[3 \times 3 \times N_m]$ and $[3 \times 3 \times N_n]$, respectively. Then, the fused feature map f_f is equal to $(\lambda \cdot f_s) \circ f_d$, and the size of f_f is $[3 \times 3 \times (N_m + N_n)]$, where $\circ$ means concatenation. As an important coefficient, the value of λ balances the weights of two features in SAR ATR. Therefore, λ is specially discussed in Section IV. Finally, the fused feature map will pass through a convolutional layer to make more comprehensive use of both scattering and deep features. By optimizing the loss function in formula (4), the feature fusion learning framework will learn the posterior distribution of target and label information to recognize SAR targets precisely.

D. Architecture and Discussion

In this work, we take the lead to treat the ASCs as set-type data rather than multiple individual scattering points. Therefore, the connection and interaction among ASCs can be extracted to represent the intrinsic scattering feature of targets. There are several existing ATR algorithms designed for set-data: Deep sets [46], graph neural network, Point Net [49], [50], set transformer [51], and so on. Considering both the local-level feature extraction capability and computational complexity, we construct the scattering feature extraction module based on PointNet++. The local and global subsets are newly constructed to represent both component- and target-level structures of targets. Besides, the twice scattering feature extraction method is proposed in this article to learn the ASC neighborhoods details and permutation-invariant features of targets. It is worth noting that any suitable set-learning algorithms beyond the mentioned networks can also be utilized to learn scattering features in SDF-Net.

The parameter settings in our framework will be introduced in the following. In the ASC extraction stage, we set the number of ASCs N to be 40 to balance the accuracy and time complexity. In the multiregion multiscale subsets' construction stage, we set three groups of subsets parameters to represent both component- and target-level structures: $[N_c = 20, N_1 = 10, R_1 = 5], [N_c = 20, N_2 = 5, R_2 = 2], [N_c = 1, N_3 = 40, R_3 = \infty]$. The channel numbers of 1×1 convolution in two feature extension modules are set to $[16, 32, 64]$ and $[128, 512]$, respectively. In the deep feature extraction module, the size of SAR image is cropped into 88×88. The four convolutional filters are with size of $[5 \times 5 \times 16]$, $[5 \times 5 \times 32]$, $[6 \times 6 \times 128]$, $[5 \times 5 \times 128]$, respectively. The filter size in the feature fusion convolutional layer is set to $[3 \times 3]$ to obtain the label vector of the target.

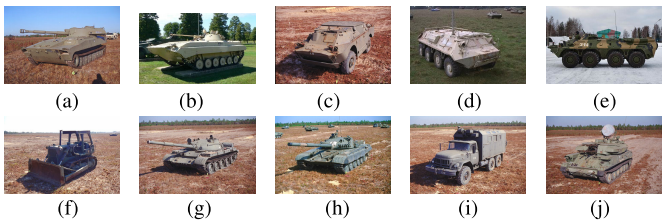


Fig. 7. Ten types of military targets in the MSTAR database in the optical format. (a) 2S1. (b) BMP2. (c) BRDM2. (d) BTR60. (e) BTR70. (f) D7. (g) T62. (h) T72. (i) ZIL131. (j) ZUS234.

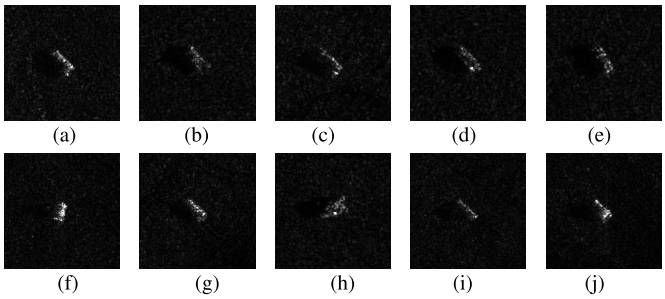


Fig. 8. Ten types of military targets in the MSTAR database in the SAR format. (a) 2S1. (b) BMP2. (c) BRDM2. (d) BTR60. (e) BTR70. (f) D7. (g) T62. (h) T72. (i) ZIL131. (j) ZUS234.

IV. Experiments and Analysis

In this section, extensive experiments will be conducted to validate the effectiveness and rationality of the proposed SDF-Net. First, the database used for experiments and five operation conditions will be introduced. Then, the experiments on hyperparameters and ablation will be set up to select suitable settings for our method and evaluate their effectiveness. Finally, the proposed SDF-Net will compare against several advanced methods to demonstrate its superiority.

A. Database and Operation Conditions' Description

The moving and stationary target acquisition and recognition (MSTAR) [52] database consists of ten types of military targets, including armored transport carriers, infantry fighting vehicles, tanks, and so on. Figs. 7 and 8 present the ten types of targets in optical and SAR format, respectively. Among them, there are various kinds of data acquired in different observation conditions. The standard operating conditions (SOC) and extended operating conditions (EOC) are constructed according to the difference in observation conditions in this article.

In the SOC, the training and testing data are observed in similar configurations, with only slight differences in depression angles. Specifically, the depression angle of training data in SOC is 17°, while that of testing data is 15°. The serial numbers and their corresponding amount of data in SOC are presented in Table I.

In the EOCs, there are four specific configurations of depression angle differences, version variants, and noise corruptions. In the EOC-1, three types of targets observed in 17, 30, and 45 depression angles are selected; among them, the data of 17° depression angle are utilized to be training data,

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TABLE I
DATA INFORMATION IN SOC

		Training Set		Testing Set	
Class	Serial Number	Depression	Number	Depression	Number
T72	132	17°	232	15°	196
2S1	B01	17°	299	15°	274
BMP2	9563	17°	233	15°	195
BRDM2	E71	17°	298	15°	274
BTR60	K10YT7532	17°	256	15°	195
BTR70	C71	17°	233	15°	196
D7	92V13105	17°	299	15°	274
T62	A51	17°	299	15°	273
ZIL131	E12	17°	299	15°	274
ZSU234	D08	17°	299	15°	274

TABLE II
DATA INFORMATION IN EOC-1 (THREE LARGE DEPRESSION VARIANT ANGLES)

		Training Set	Testing Set	
Class	Serial Number	17°	30°	45°
2S1	B01	299	288	303
BRDM2	E71	298	287	303
ZSU234	D08	299	288	303

TABLE III
DATA INFORMATION IN EOC-2 (TWO LARGE DEPRESSION VARIANT ANGLES)

		Training Set		Testing Set	
Class	Serial Number	Depression	Number	Depression	Number
T72	A64	17°	299	30°	288
2S1	B01	17°	299	30°	288
BRDM2	E71	17°	298	30°	287
ZSU234	D08	17°	299	30°	288

and the other two categories of data with depression angles of 30° and 45° are used to test the generalization performance of the network, while, in EOC-2, there are four types of targets, and the depression angles of training data and testing data are 17° and 30°, respectively. In EOC-3, there are large version variants between training data and test data. Finally, five additive Gaussian noises are manually added to make the SNR of the image of 10, 5, 0, −5, and −10 dB to simulate the real-world noise corruption situation in EOC-4. The huge variants in the EOCs would pose severe challenges in SAR ATR since the characteristics of SAR data are sensitive to the observation angles, target versions, and noise corruption. The serial numbers and their corresponding amount of data in EOC-1, -2, and -3 are presented in Tables II–IV, respectively. The detailed information of EOC-4 is the same as that of SOC in Table I.

B. Experiments on Hyperparameters

There are two key hyperparameters in the proposed feature fusion framework: the weight coefficient λ and the sampling scale S. The weight coefficient λ balances the scattering and deep features, while the sampling scale S determines physical parameters and structure details in scattering features by the amount of central and neighborhood ASCs. Therefore, these two hyperparameters have a great impact on the performance of the network and should be discussed technically.

TABLE IV
DATA INFORMATION IN EOC-3 (VERSION VARIANTS)

	Class	Serial No.	Depression	No.
Training Set	T72	132	17°	232
	BMP2	9563	17°	233
	BRDM2	E71	17°	298
	BTR70	C71	17°	233
Testing Set	T72	812	17°,15°	426
		A04	17°,15°	573
		A05	17°,15°	573
		A07	17°,15°	573
		A10	17°,15°	567
	BMP2	9566	17°,15°	428
		C21	17°,15°	429

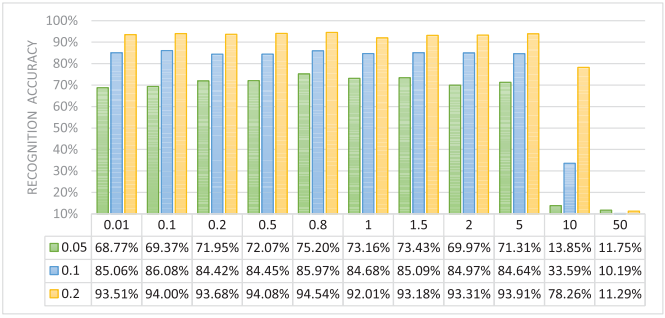


Fig. 9. Recognition accuracy on a different hyperparameter λ.

1) Experiments on λ: As for λ, we set the range of it to be $\lambda \in [0.01, 0.1, 0.2, 0.5, 0.8, 1, 1.5, 2, 5, 10, 50]$ to explore its influence on the feature fusion. To rule out other factors, we set the sampling scale S as introduced in Section III and conduct experiments under sampling rate of $\mu = 0.05$, $\mu = 0.1$, $\mu = 0.2$, respectively. The experiment results are shown in Fig. 9. As can be seen, the difference between the maximum and the second-highest recognition accuracy is 1.77% when μ is set to 0.05, while the difference decrease to 0.46% when $\mu = 0.2$. The difference in accuracy decline shows that the recognition accuracy is greatly affected by λ, especially when μ is small.

When λ is set to 0.01, the proportion of scattering features is negligible, and the fusion framework is close to the deep feature extraction network; thus, the recognition accuracy is not very high. However, the fusion network will overfocus on the scattering feature when λ is set to 50, resulting in a significant decrease in accuracies of 63.45%, 75.78%, and 83.25%, respectively, compared to the best results under three sampling proportions. It is obvious that the network cannot balance these two features properly if its value is too large or too small. Concluded from the comparison results, though the accuracy of $\mu = 0.1, \lambda = 0.1$ is 0.11% slight higher than that of $\mu = 0.1, \lambda = 0.8$, the fusion network can achieve better performance when λ is set to 0.8 than other λ mostly. Therefore, we set λ to be 0.8 in the following experiments to fuse the two features more complementary and robustly.

2) Experiments on S: The sampling scale S in scattering feature extraction module indicates the structures and regions in which the model can extract. We set up an experiment

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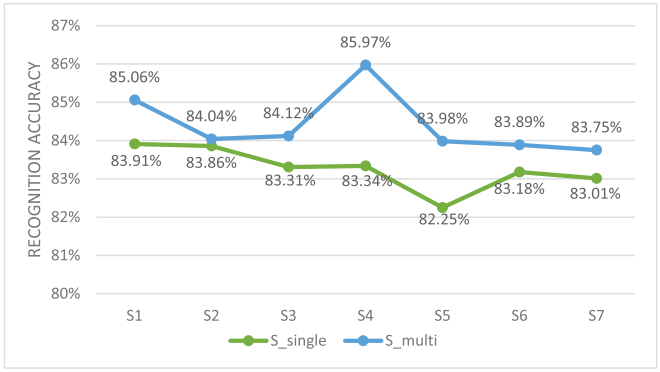


Fig. 10. Recognition accuracy on different scales S.

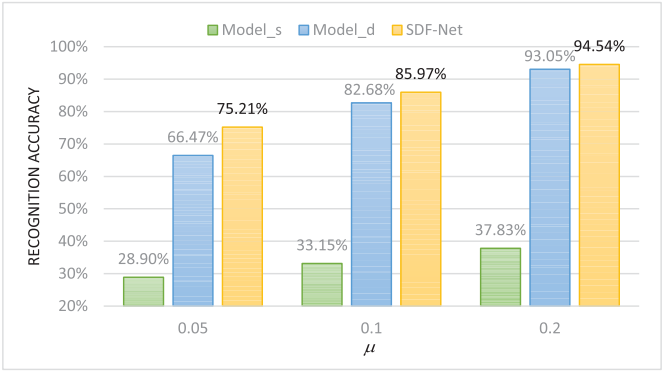


Fig. 11. Comparison of the scattering feature extraction model Model$_s$, the deep feature extraction model Model$_d$, and the feature fusion model SDF-Net on recognition accuracy.

on S in order to exploit useful features in single-scale to the most extent, where S is set to values in the following ranges: S_{single}: $\{S_{\text{single1}}$: $[N_1 = 20, N_1 = 10, R_1 = 5]$, S_{single2}: $[N_c = 20, N_2 = 5, R_2 = 2]$, S_{single3}: $[N_c = 15, N_3 = 10, R_3 = 5]$, S_{single4}: $[N_c = 15, N_4 = 5, R_4 = 2]$, S_{single5}: $[N_c = 10, N_5 = 10, R_5 = 5]$, S_{single6}: $[N_c = 10, N_6 = 5, R_6 = 2]$, S_{single7}: $[N_c = 1, N_7 = 40, R_7 = \infty]\}$. Among all the single scales, the first six refer to regions and neighborhoods of different sizes, while the last one represents the whole structure of the target. Similarly, the sampling rate is set to $\mu = 0.1$, and the weight coefficient λ is set to 0.8. The recognition results of a different single scale S_{single} are shown in the green line in Fig. 10, where the abscissa $S_{\text{i}}(i \in [1, 7])$ of the green nodes correspond to single-scale configuration $S_{\text{singlei}}(i \in [1, 7])$. Observed from the results that the fusion model achieves the best performance when $S_{\text{single}} = S_1$ and the worst performance when $S_{single} = S_5$, the difference in accuracy between them reaches up to 1.66%, which reflects the effect of regions and target structures in various scales on scattering features.

Besides, in order to exploit the effect of multiscale scattering features on the feature fusion framework, we select several configurations with highest recognition accuracy from the single scale ones to construct multiscale experiments: S_{multi}: $\{S_{\text{multi1}}$: $[S_{\text{single1}} + S_{\text{single2}} + S_{\text{single4}}]$, S_{multi2}: $[S_{\text{single1}} + S_{\text{single2}} + S_{\text{single3}}]$, S_{multi3}: $[S_{\text{single1}} + S_{\text{single2}} + S_{\text{single6}}]$, S_{multi4}: $[S_{\text{single1}} + S_{\text{single2}} + S_{\text{single7}}]$, S_{multi5}: $[S_{\text{single1}} + S_{\text{single3}} + S_{\text{single4}}]$, S_{multi6}: $[S_{\text{single1}} + S_{\text{single4}} + S_{\text{single6}}]$, S_{multi7}: $[S_{\text{single3}} + S_{\text{single4}} + S_{\text{single6}}]\}$. The setups of other parameters are all the same as those in single-scale experiments. As shown in the blue line in Fig. 10, the recognition accuracy of multiscale configurations is mostly higher than that of single-scales. Besides, the feature fusion learning framework SDF-Net achieves the best recognition accuracy with 85.97% in S_{multi4}. Look back to the configurations of multiscales, though S_{multi1} integrates the top three single scales with highest accuracy, S_{multi4} contains both the single scale with best accuracy and S_{single7}, which represents all of ASCs globally. This indicates that the combination of local and global scattering features in ASCs is able to exploit multiple scales of information to the best extent and, thus, has the best performance on recognition. Therefore, we set S to be S_{multi7} in the following experiments to better mine the multiscale scattering feature.

C. Ablation Experiments

In order to prove the effectiveness of the proposed SDF-Net and the complementary between scattering and deep features, we set up two ablation experiments on the two kinds of features, respectively. Specifically, the proposed SDF-Net is divided into two parts: the multiregion multiscale scattering feature extraction model and the deep feature extraction model. The structures of these two networks are all exactly the same as those in the original fusion model, just without the other part. The two networks are indicated as Model$_s$ and Model$_d$ for convenience. In order to keep other factors consistent in experiments, we set $\lambda = 0.8$ and scale S as S_{multi4} introduced above. The experiments are carried out under configurations of $\mu = 0.05$, $\mu = 0.1$ and $\mu = 0.2$, respectively, and the recognition results are presented in Fig. 11.

The ablation experiment results demonstrate that both the scattering feature extraction model and the deep feature extraction model have a decrease in recognition accuracy without the other one. As for the former model, the recognition accuracy decreases by 46.31%, 52.82%, and 56.71% with sampling proportions of $\mu = 0.05$, $\mu = 0.1$, and $\mu = 0.2$, respectively. Specifically, the scattering network Model$_s$ achieves recognition accuracy of no more than 40%; it can be concluded that the scattering feature is not so sufficient for the training of the network due to the loss of contour, key point, and other kinds of information in the process of ASC extraction. On the other hand, the deep feature extraction model decreases by 8.74%, 3.29%, and 1.49% in recognition accuracy compared to the feature fusion model under the three sampling proportions, which demonstrates that the recognition performance is already precise to a certain degree with only the deep feature extraction model. The lower recognition accuracy proves from the side that the scattering feature can provide complementary physical information to the deep feature.

Moreover, the recognition accuracy difference between the deep feature extraction model and SDF-Net is getting smaller with the increase of sampling proportion μ. This indicates that the generalization performance of the deep

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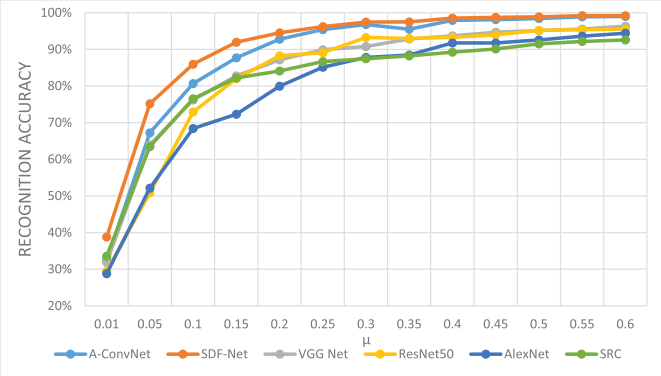


Fig. 12. Comparison results of recognition accuracy on SOC.

feature extraction model enhances with the increase in the training data, and meanwhile, the auxiliary effect of scattering features decreases. Therefore, we will prove the validity of the proposed feature fusion learning framework SDF-Net under complex observation conditions in the following comparative experiments.

D. Framework Comparison

To validate the superiority of the proposed feature fusion learning framework quantitatively, we set up comparison experiments between SDF-Net and several competitive ATR algorithms under conditions of limited data, depression angle differences, target version differences, and noise corruption in this section.

1) Results Under SOC: The ASC sets reflect structures of targets at the component level, which means that scattering features represent the spatial topological structures of targets and provide additional electromagnetic scattering information beyond discriminant features learned by deep networks. Thus, the extra scattering information can be exploited to improve the recognition ability in SAR ATR. In order to verify this property, we compare SDF-Net to five benchmark methods with limited data, where the proportion of training samples ranges from 1% to 60%. The five typical ATR algorithms for comparison are A-ConvNet [2], VGG Net [34], ResNet50-V2 [53], AlexNet [54], and sparse representation classifier (SRC), respectively. Among all the algorithms, the former four are famed deep learning models that have significant performance on ATR tasks, while the last one is a widely used traditional ATR algorithm. The comparison experiment results and detailed values of recognition accuracy under SOC are shown in Fig. 12 and Table V, respectively. The confusion matrix of SDF-Net with all training samples is summarized in Table VI. In addition, in order to prove the robustness of SDF-Net, we set up an experiment to show the difference between the recognition accuracy declines of models as the proportion of training data decreases. The comparison results are shown in Fig. 13.

It is intuitive from the comparison results that SDF-Net achieves the highest recognition accuracy at all sampling proportions. Besides, the accuracy at $\mu = 0.05$ of SDF-Net is 7.98% higher than that of A-ConvNet, which is the baseline

TABLE V
DETAILED RECOGNITION ACCURACY (%) OF SAR ATR ALGORITHMS ON SOC

μ	SDF-Net	A-ConvNet	VGG Net	ResNet50	AlexNet	SRC
0.01	**38.79**	32.49	31.85	29.53	28.79	33.57
0.05	**75.21**	67.23	63.97	50.87	52.16	63.49
0.1	**85.97**	80.70	76.23	72.95	68.42	76.51
0.15	**91.95**	87.73	82.81	82.04	72.32	82.23
0.2	**94.54**	92.78	87.13	88.30	79.97	84.13
0.25	**96.23**	95.41	89.91	89.12	85.15	86.69
0.3	**97.45**	96.81	90.81	93.28	87.79	87.46
0.35	**97.54**	95.51	92.82	93.02	88.47	88.25
0.4	**98.55**	97.94	93.73	93.33	91.75	89.23
0.45	**98.74**	98.11	94.68	93.96	91.78	90.12
0.5	**98.88**	98.42	95.14	95.22	92.60	91.51
0.55	**99.20**	98.88	95.57	95.38	93.61	92.17
0.6	**99.23**	99.00	96.34	95.60	94.42	92.58

TABLE VI
CONFUSION MATRIX OF SDF-NET UNDER SOC

Class	2S1	BMP2	BRDM2	BTR60	BTR70	D7	T62	T72	ZIL131	ZSU234
2S1	274	0	0	0	0	0	0	0	0	0
BMP2	0	194	0	1	0	0	0	0	0	0
BRDM2	0	0	274	0	0	0	0	0	0	0
BTR60	1	0	0	193	0	1	0	0	0	0
BTR70	1	0	0	0	192	0	2	0	1	0
D7	0	0	2	0	0	269	0	2	1	0
T62	0	1	0	0	1	0	270	0	1	0
T72	0	0	2	0	1	1	0	192	0	0
ZIL131	0	0	0	0	0	0	0	0	274	0
ZSU234	0	0	0	1	0	0	0	0	0	273

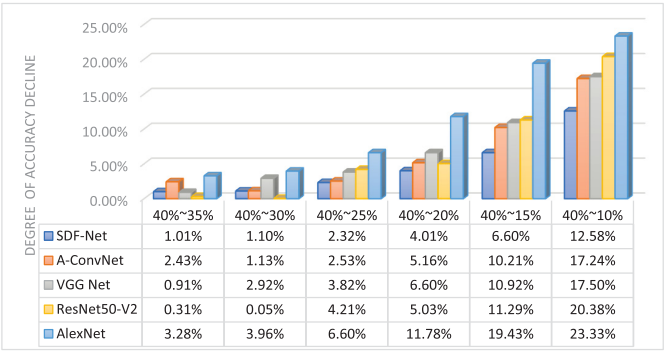


Fig. 13. Comparison of recognition accuracy decline with the decrease in sampling proportion.

model in SAR ATR, while, thanks to the fusion between scattering features and deep features, SDF-Net is able to outperform other networks, especially in the case of fewer training samples, which demonstrates that the scattering features can provide additional supervised information of targets especially when the samples are insufficient. In addition, the confusion matrix of SDF-Net shows that almost all of the samples are correctly recognized under SOC. When it comes to Fig. 13, the recognition accuracy declines to maintain an upward tendency as the reduction in the number of data increases. In combination with both recognition accuracy and the decrease in accuracy, although the accuracy decrease in ResNet50-V2 is slightly lower than that of SDF-Net, the recognition accuracy of the latter is much higher than all the other typical networks. Besides, the recognition accuracy drops of SDF-Net are all lower than the other models when the sampling proportion reduction is more than 5%, which indicates the robustness of SDF-Net.

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TABLE VII
COMPARISON RESULTS UNDER SOC

	Model	Sampling Proportion	Recognition Accuracy
ASC-based Model	**SDF-Net**	**60%**	**99.23%**
	SDF-Net	**100%**	**99.58%**
	FEC	100%	99.27%
	FGL	100%	99.08%
	DA-ASC	100%	99.48%
Non-Model-based Network	CAE-HL-CNN	72.81%	97.86%
	A-ConvNet	100%	99.26%
	RLM	100%	97.10%
	AMCNN	100%	99.35%
	SECNN	100%	99.18%

Finally, the superiority of multilevel scattering features utilized in SDF-Net and the effectiveness of the fusion strategy are verified. In order to explore how few data can use SDF-Net to achieve the effect of the same level as other methods using more training data, we set up an experiment in which the recognition accuracies between SDF-Net with 60% and 100% training data and other methods with more training data are compared. Besides the typical deep networks, several advanced SAR ATR algorithms are selected to compare against SDF-Net to demonstrate its superiority in recognition ability. Among all the comparison methods, FEC [33], FGL [32], and data augmentation based on ASC (DA-ASC) [27] are the up-to-date ASC-based SAR ATR algorithms. CAE-HL-CNN [55] and the ranking loss module (RLM) [56] are designed for SAR ATR with limited training data. Attention mechanism CNN (AMCNN) [57] and squeeze and excitation network (SECNN) [58] are non-ASC-based SAR ATR algorithms proposed in recent years. It is worth noting that FEC is the first model to fuse scattering and deep features at the feature level, while FGL takes both global and local ASC features into account. Therefore, the comparison with them can better demonstrate the advantages of the learning method for multilevel scattering features in SDF-Net. The comparison with other non-ASC-based models also verifies the superiority of the feature fusion strategy. The detailed comparison results are shown in Table VII.

The comparison results in Table VII show that SDF-Net can use the same amount or even fewer data to perform better than both the ASC-based models and nonmodel-based networks. SDF-Net achieves 99.23% recognition accuracy with only 60% training samples under SOC, which is already equal to or even better than most of the comparison methods using 72.81% or 100% samples. When it comes to all training samples, the recognition accuracy of SDF-Net reaches up to 99.58%, which is competitive with both of the ASC-based models and nonmodel-based networks. The results indicate that the proposed multiregion multiscale scattering feature and deep feature fusion learning framework can better exploit the electromagnetic information of SAR data and ASCs than the other ASC-based models with limited data. Besides, the fusion strategy is effective to extract more critical supervision information for SAR ATR by comparing the SDF-Net against the nonmodel-based networks.

TABLE VIII
COMPARISON RESULTS UNDER EOC-1

	Model	Sampling Proportion	Recognition Accuracy /30°	Recognition Accuracy /45°
ASC-based Model	**SDF-Net**	**60%**	**98.53%**	**85.27%**
	SDF-Net	**100%**	**99.20%**	**86.57%**
	FEC	100%	99.19%	81.08%
	FGL	100%	99.07%	83.39%
	DA-ASC	100%	98.61%	74.48%
Non-Model-based Network	AConvNet	60%	97.61%	80.87%
	AConvNet	100%	98.62%	84.45%

TABLE IX
COMPARISON RESULTS UNDER EOC-2

Model	Sampling Proportion	Recognition Accuracy
SDF-Net	**30%**	**92.53%**
A-Convnet	30%	91.45%
RotANet	40%	90.35%
SDF-Net	**60%**	**96.31%**
CAE-HL-CNN	66.95%	91.15%
SDF-Net	**100%**	**97.16%**
RLM	100%	86.70%

2) Results Under EOC-1 and EOC-2: We set up two extended operation conditions EOC-1 and EOC-2 to verify the superiority and robustness of the proposed SDF-Net in the case of huge depression angle differences. In EOC-1, the ASC-based SAR ATR algorithms applied in the case of depression angle differences, including FEC, FGL, DA-ASC, and typical SAR ATR algorithm A-ConvNet, are selected to compare against SDF-Net. We test the recognition accuracy of SDF-Net with 60% and 100% training samples to compare against ASC-based models with 100% data to verify the rationality and effectiveness of the ASC scattering feature extraction method in SDF-Net. In EOC-2, SAR ATR algorithms applied in the case of depression angle differences, including RotANet, CAE-HL-CNN, and RLM, are selected to compare against SDF-Net. Similarly, we test the recognition accuracy of SDF-Net with the same or even less data sampling proportion than that of the comparison methods to verify the rationality of scattering features and the effectiveness of fusion strategy in our method. The comparison results under EOC-1 and EOC-2 are shown in Tables VIII and IX, respectively. The confusion matrices of SDF-Net with all training samples under EOC-1 and EOC-2 are shown in Tables X and XI, respectively.

In the EOC-1, SDF-Net reaches the recognition accuracy up to 98.53% with only 60% training samples in the case of 30° test data, which is already higher than DA-ASC and A-ConvNet with 100% data. When it comes to 100% training samples, the recognition accuracy of SDF-Net is 99.20%, while the highest recognition accuracy of other comparison algorithms is 99.19%. In the case of 45° test data, SDF-Net achieves 85.27% recognition accuracy with only 60% training

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TABLE X
CONFUSION MATRIX UNDER EOC-1

Depression Angle	Class	2S1	BRDM2	ZSU234
30°	2S1	285	2	1
	BRDM2	5	281	1
	ZSU234	0	2	286
45°	2S1	258	26	19
	BRDM2	25	263	15
	ZSU234	30	16	257

TABLE XI
CONFUSION MATRIX UNDER EOC-2

Class	T72	2S1	BRDM2	ZSU234
T72	285	1	2	0
2S1	3	282	2	1
BRDM2	3	5	276	3
ZSU234	5	6	2	275

TABLE XII
COMPARISON RESULTS UNDER EOC-3

	Model	Sampling Proportion	Recognition Accuracy
ASC-based Model	**SDF-Net**	**40%**	**93.69%**
	SDF-Net	**100%**	**98.76%**
	FEC	100%	98.48%
	DA-ASC	100%	98.80%
Non-Model-based Network	RotANet	40%	92.96%
	A-Convnet	100%	97.93%

data, which is higher than all the comparison methods with 100% training data. In the EOC-2 comparison experiment, SDF-Net achieved better recognition accuracy in all three data proportion tests. SDF-Net reached up to 92.53%, 96.31%, and 97.16% recognition accuracies with 30%, 60%, and 100% training data, while the comparison algorithms all got lower accuracy with the same or higher sampling proportion. The comparison results demonstrate that SDF-Net can use fewer data to achieve the same or even better recognition performance compared to other methods, which proves that the SDF-Net can utilize both the scattering features and deep features more effectively in SAR ATR with huge depression angle differences. The confusion matrix under EOC-1 and EOC-2 shows that the test samples are almost evenly classified.

3) Results Under EOC-3: Comparison experiments with different version variants are conducted in order to further demonstrate the robustness of SDF-Net. ASC-based models, namely, FEC and DA-ASC, and nonmodel-based networks, namely, RotANet and A-ConvNet, are selected to compare with SDF-Net. The detailed comparison results are shown in Table XII. The recognition accuracy of SDF-Net with 40% training samples is 93.69%, which is higher than that of RotANet. When it comes to 100% data, the recognition

TABLE XIII
CONFUSION MATRIX UNDER EOC-3

Class	Serial Number	BMP2	BRDM2	BTR70	T72
BMP2	9566	426	0	2	0
	c21	421	3	5	0
T72	812	3	5	0	418
	A04	0	2	3	568
	A05	2	2	3	566
	A07	1	3	5	564
	A10	1	0	1	565

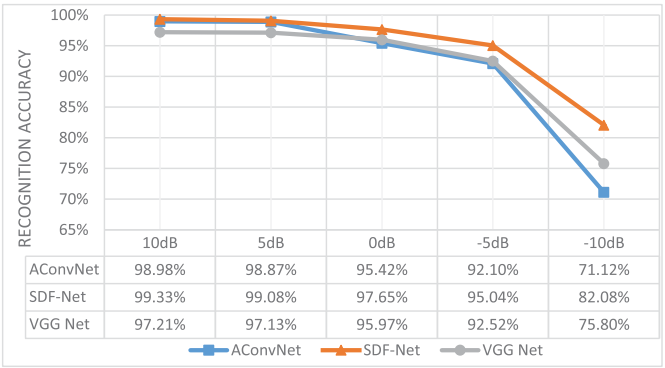


Fig. 14. Comparison of recognition accuracy under EOC-4.

TABLE XIV
CONFUSION MATRIX WITH −10-dB SNR UNDER EOC-4

Class	2S1	BMP2	BRDM2	BTR60	BTR70	D7	T62	T72	ZIL131	ZUS134
2S1	232	4	6	5	5	6	4	3	3	6
BMP2	5	163	3	2	4	3	7	3	2	3
BRDM2	9	8	226	5	6	4	7	2	5	2
BTR60	2	2	8	159	6	4	2	7	4	1
BTR70	5	3	8	6	153	5	7	4	3	2
D7	3	4	5	5	3	237	6	2	4	5
T62	5	9	5	4	3	5	228	3	6	5
T72	4	2	5	3	4	2	5	166	2	3
ZIL131	2	6	5	4	5	7	6	6	226	7
ZSU234	6	5	7	5	4	2	7	5	5	228

performance of SDF-Net is competitive with the comparison methods. The confusion matrix of EOC-3 in Table XIII shows that the test samples are almost evenly correctly recognized by SDF-Net. The comparison results of EOC-3 demonstrate that the SDF-Net can better mine useful supervision knowledge to enhance the recognition ability in the condition of version variants.

4) Results Under EOC-4: Comparison experiments with five SNR-level data of 10, 5, 0, −5, and −10 dB are conducted. Typical SAR ATR algorithm A-ConvNet and deep model ResNet50-V2 are selected to compare against SDF-Net. The detailed comparison results are shown in Fig. 14. The confusion matrix with −10-dB SNR under EOC-4 is shown in Table XIV. The SDF-Net gets the best recognition accuracy at all five noise levels. The recognition accuracy decrease in SDF-Net is also the smallest when SNR decreases from 10 to −10 dB. The confusion matrix shows that the SDF-Net can recognize testing samples evenly even in the case of −10-dB noise. The comparison results demonstrate that

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SDF-Net can achieve robust and effective performance compared to both the typical SAR ATR and optical ATR methods in the case of noise corruption.

V. Conclusion

A novel end-to-end feature fusion learning framework SDF-Net based on scattering feature and deep feature has been proposed for SAR ATR in this article. We characterize the ASCs as set-type data instead of individual scattering points, and then, the framework can exploit the connection and interaction among them as local and global permutation-invariant scattering features that are extracted at the component level. The ASCs located in multiregion and multiscale can represent the SAR target structures in a physical semantic manner, which enhances the interpretability of SDF-Net. Finally, the scattering feature and deep feature are fused with a weight coefficient to make the combination of these two features more complementary. This kind of fusion strategy improves the generalization ability of the proposed feature fusion learning framework, especially in the case of huge image modality discrepancies. Extensive experiments under complex observation conditions verify the efficiency of the proposed SDF-Net. The recognition accuracy of SDF-Net is much higher than that of the benchmark network in the case of both SOC and EOCs. In future work, we will further exploit the utilization of essential attributes of SAR data in different feature spaces to improve the performance of SAR ATR.

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